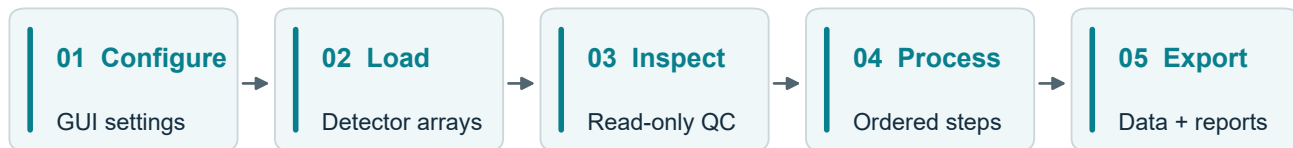


a Batch workflow



b Fixed numerical operation order

- | | | | | | |
|----|--------------------|----|-----------------------|----|---------------------|
| 01 | Dark subtraction | 06 | Mask | 11 | Rotate / flip |
| 02 | Flat correction | 07 | Absolute clipping | 12 | Block-mean binning |
| 03 | Background offset | 08 | Percentile clipping | 13 | Intensity transform |
| 04 | Validate clipping | 09 | Negative clipping | 14 | Gamma |
| 05 | Region of interest | 10 | Hot-pixel suppression | 15 | Normalization |

c Independent tools

CBF zero-value repair

Target mask → write → verify
Configured zeros; not saturation recovery

Ideal planar geometry

Wavelength + detector geometry
Q, 2θ , radius and d-spacing conversion

Preprocessing and QC precede downstream integration or fitting.